EE-559
Deep Learning

What's on today?

- Learning outcomes (recap!)
- Project deliverables
- Assessment criteria
- Poster and poster session
- Paper
- Convergence to project success!

Learning outcomes

EE-559: Learning outcomes

- By the end of the course, you will be able to:
  - **Justify** the choices for training and testing a deep learning model
  - **Interpret** the performance of a deep learning model
  - **Analyze** the limitations of a deep learning model
  - **Propose** new solutions for a given application

Deliverables

EE-559: Group mini-project deliverables

- The code + a screencast of the code running
- A poster (*template provided*)
- A 3-page paper (*template provided*)

Assessment criteria

Group mini-project

- **Assessment criteria**
  - Literature review and methodology (up to 20 marks)
  - Evaluation, testing, and analysis (up to 20 marks)
  - Communication of the findings (up to 20 marks)

Group mini-project: assessment criteria (1/3)

- **Literature review and methodology**
  - thoroughness of literature review
  - clarity of the objectives and problem definition
  - evidence of creativity and novelty of the adopted methodology

Group mini-project: assessment criteria (2/3)

- **Evaluation, testing, and analysis**
  - quality of the results of the proposed solution and their analysis
  - discussion of the limitations of the proposed solution
  - justification of the choices for the experiments
  - evidence of critical thinking

Group mini-project: assessment criteria (3/3)

- **Communication of the findings**
  - ability to clearly communicate the findings
    - in the poster
    - during the oral presentation of the poster
    - in the written report

Poster

Poster preparation

- Highlight your key **results** and **findings**
- Facilitate visual impact: use **diagrams** and **plots**
- Prioritize clarity and conciseness
  - keep the poster **focused** and **uncluttered**
  - **do not** copy paste & paste material from your paper!

Poster

EPFL

Project Title

MINTS Multimodal Intelligent Systems

Name_Surname1, Name_Surname2, Name_Surname3

Group GroupNumber

Problem definition

- Define the problem you have addressed with this project
- Note: do **not** just copy-paste text from your report.

Validation

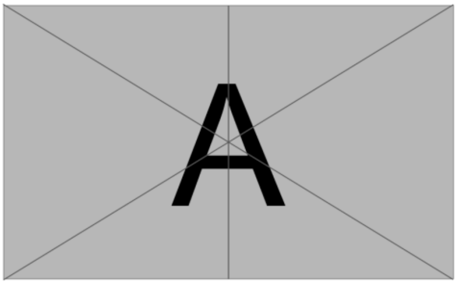
- Present the Evaluation, Testing and Analysis here.
- Include add plots, tables with results, present the performance measures

Dataset	Naive	Flexible	Better?
CLEVELAND	83.3 ± 0.6	80.0 ± 0.6	×
GLASS2	61.9 ± 1.4	83.8 ± 0.7	✓
CREDIT	74.8 ± 0.5	78.3 ± 0.6	

Related works

- Discuss your key references

Method
- Present your method
- Use diagrams and/or formulas to explain your method



Dataset(s)
- List the dataset(s) you used

Limitations
- Discuss the limitations of your work

Conclusion
- Present your conclusions and possible future work

References
Add your **three** key references here. Use IEEE referencing style. For example,
[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," Advances in neural information processing systems, vol. 30, 2017.
[2]
[3]

EE-559: Deep Learning, 2026

Project Title

Name_Surname1, Name_Surname2, Name_Surname3
Group GroupNumber

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[3]

EE-559: Deep Learning, 2026

Poster presentation

- All group members to be involved in the presentation
- Engage with the questions
- Keep your answers short and to the point
- **Enjoy the experience!**

Poster session on 27th May

- Open session
- Valuable opportunity to showcase your work and get feedback!
- At least one group member should be at the poster for the whole poster session
- All members to be present during the group assessment slot

Paper

About the paper

- Clearly present the objective(s) of the project
- Identify and discuss relevant related works that is specific to your project
- Leave enough space for the analysis of the results
- Support *all* your statements with evidence (from papers or your results)
- Use consistently the IEEE references style (see *template*)
- Remain within the 3-page limit (including references)!
- Proof-read carefully before submission!

Project Title

Name_Surname2 Name_Surname2 Name_Surname3

Group GroupNumber

Abstract

Add your abstract here. Mention the issue(s) you have addressed, why they are important, and describe your proposed solution. Do not edit the style of this document (e.g., font size, margins) and do not to exceed the 3-page limit.

Keywords: add your keywords here.

1. Introduction

Add your introduction here. Select a current limitation or a relevant new problem: identify a specific problem you want to address. Clarify the objective(s) and the problem definition. State any hypotheses you made and reference the sources you used. Some common LaTeXcommands are listed below.

Section: you can refer to a section as Sec. 2.

Equation:

$$x + y = 0. \tag{1}$$

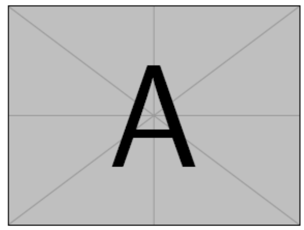


Figure 1. Insert your caption here.

DATA SET	NAIVE	FLEXIBLE	BETTER?
CLEVELAND	83.3± 0.6	80.0± 0.6	×
GLASS2	61.9± 1.4	83.8± 0.7	√
CREDIT	74.8± 0.5	78.3± 0.6	

Table 1. INSERT YOUR CAPTION HERE.

You can refer to an equation as Equation 1.

Figure: you can refer to a figure as Fig. 1.

Table: you can refer to a table as Tab. 1 in your text. https://www.tablesgenerator.com/ is useful for creating custom tables.

Reference: you can cite a source using command `\cite`, e.g. [1]. To add a reference, you can find your source on Google Scholar, click "Cite" and select BibTeX. Then copy the reference to `main.bib`.

2. Related Work

Add your literature review here. Discuss the limitations of the literature.

3. Method

Use what you have learnt in EE-559 to address the limitations you identified. Describe and motivate your methodology.

4. Validation

Implement your ideas and test them. Add your evaluation, testing and analysis here. Justify the choices for the experiments. Analyse the results and the performance: why does your hypothesis work / doesn't work? Compare with alternative ideas / hypotheses. Discuss the limitations of the proposed solution.

5. Conclusion

Add your conclusion here.

References

[1] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," *Advances in neural information processing systems*, vol. 30, 2017.

Project objective(s)

- What is the need?
  e.g. "Scarcity of diverse, high-quality labelled data for hateful speech in text [REF1, REF2]."

- What is the goal?
  e.g. "To overcome the lack of the high-quality data for the hateful speech in the text domain."

- What did we do?
  e.g. "*We used X [REF1] in combination with Y [REF2] to enhance data availability and improve detection accuracy."*

Literature review

- What are the closest related works to your project?
  Identify the most relevant papers that align with your research goals or solve a similar problem.

- What are their limitations?
  Discuss the gaps or weaknesses in those papers. State which gaps/weaknesses you addressed.

- How does your project differ and (in which aspects) is it better?
  Explain how your project improves upon these papers, and what are the differences.

Methodology

- What are the papers that informed your approach?
  Briefly discuss the paper(s) whose methodology inspired your approach to the project.
- What was done by you?
  Clearly separate what is your actual work from what was done in the literature.
- Which models and datasets did you use?
  Justify the choice of the models (e.g. BERT, Llama3.2, Qwen3.5-Omni) and dataset(s) you used.

Evaluation, testing, and analysis (1/2)

- Clearly explain your experimental setup, including any pre-processing steps.
- Use baselines for comparison.
- Ensure that the setup for the comparison with the literature is fair.
- Use multiple seeds and evaluate the variance in your model's performance.
- Make sure that the data split reflects the one that was used in the literature.

Evaluation, testing, and analysis (2/2)

- Support all your statements with a reference to the literature or your results.
- Show what happens if key elements of the method are removed (ablation study).
- Explain the possible reasons for the results you obtained.
- Acknowledge the limitations of your method.

Convergence to project success!

Your mini-project choices

To be aligned with the course learning objectives
and your broader transversal skill development

- Learning objectives
  - interpret the performance of a deep learning model
  - analyze the limitations of a deep learning model
  - justify the choices for training and testing a deep learning model
  - propose new solutions for a given application
- Transversal skill development
  - respect relevant legal guidelines and ethical codes
  - take account of the social and human dimensions
  - …

Data

Objective: high-quality, well-labelled data

Get familiar with the benchmarks on the task you are addressing
Do you really need to train a model from scratch?
Can you apply fine-tuning?
Check if your model overfits
Discuss your choices in the report

The amount of data required for training a
neural network depends on several factors,
such as problem complexity and model size.

Model

- **Multi-modal models are cool!**

  however check if they are necessary for your problem

- **You can use a pre-trained model as a basis of your project**

  justify in the report why you are using a pre-trained model

- **Having the best-performing model in its class is highly gratifying!**

  however this is not the main goal of the mini-project
  it depends on the other contributions/innovations with respect to the learning outcomes
  (performance can be slightly lower if compensated by innovation/creativity elsewhere)

Framing your mini-project

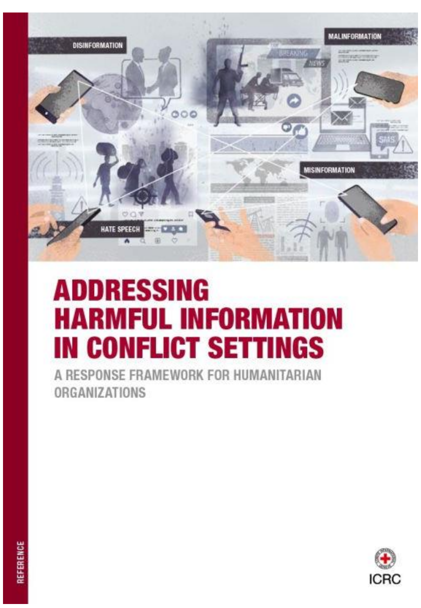
- **Select a current limitation or a relevant new problem**
  - identify a specific problem you want to address
  - discuss in the report the limitations of the literature
  - use what you have learnt in EE-559 to address the limitations you identified
- **Implement your ideas**
- **Test them**
- **Analyse the performance**
  - why does your hypothesis work / doesn't work?
  - compare with alternative ideas / hypotheses

Do not forget to …

- State in the report any hypotheses you made
- Reference the sources you use
- Comment your code

What's next?


EPFL EssentialTech Centre
https://actu.epfl.ch/news/save-the-date-peacetech-hackathon-2026
PeaceTech
PEACETECH HACKATHON 2026
MISINFORMATION, DISINFORMATION AND HATE SPEECH (MDH)
26-27 September 2026
hate
Over two days, actors from diverse academic and technical backgrounds will address real-world challenges proposed by international organizations and NGOs, creating scalable tools for social impact.
With the support of:
Fondation Schlumberger pour l'Education et la Recherche
Ville de Lausanne
EPFL Campus, Rte Cantonale, 1015 Lausanne


DISINFORMATION
MALINFORMATION
MISINFORMATION
HATE SPEECH
ADDRESSING HARMFUL INFORMATION IN CONFLICT SETTINGS
A RESPONSE FRAMEWORK FOR HUMANITARIAN ORGANIZATIONS
REFERENCE
ICRC


IFRC
World Disasters Report
2026
IFRC
Truth, Trust and Humanitarian Action in the Age of Harmful Information

SESSIONS

12th session of the Group of Independent Eminent Experts on the Implementation of the Durban Declaration and Programme of Action

DATE
05 - 08 May 2026

LOCATION
ROOM V, PALAIS DES NATIONS, GENEVA

https://webtv.un.org/en/asset/k1f/k1fs7w3dug

RESEARCH-ARTICLE | FREE ACCESS

MM-HSD: Multi-Modal Hate Speech Detection in Videos

Authors: Berta Céspedes-Sarrias, Carlos Collado-Capell, Pablo Rodenas-Ruiz, Olena Hrynenko, Andrea Cavallaro | Authors Info & Claims

MM '25: Proceedings of the 33rd ACM International Conference on Multimedia • Pages 2546 - 2555
https://doi.org/10.1145/3746027.3754558

Published: 27 October 2025 Publication History

Check for updates

RESEARCH-ARTICLE | FREE ACCESS

Specializing General-purpose LLM Embeddings for Implicit Hate Speech Detection across Datasets

Authors: Vassiliy Cheremetiev, Quang Long Ho Ngo, Chau Ying Kot, Alina Elena Baia, Andrea Cavallaro | Authors Info & Claims

DHOW '25: Proceedings of the 2nd International Workshop on Diffusion of Harmful Content on Online Web
Pages 23 - 36 • https://doi.org/10.1145/3746275.3762209

Published: 26 October 2025 Publication History

Check for updates

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